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Problem Solving

Program

- Python syntax
- Functions
- Modularity
- Data structures (list, dict, set, tuple)
- Git & Github
- Algorithms & Complexity
- Problem solving

Problem Solving

5-step problem solving method

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Restate the problem
- Proves you understand
- Catch misleading early

Identify edge cases
- Force you to think about the boundaries before getting attached to an approach

Brute force first
- Gives you a working baseline to reason from
- Don't skip it outloud

Optimize
- Now, look for patterns
- Hash map ? Sorting ? two pointers ?

Code, test, name complexity
- Write the code last, test one example by hand
- then name the big O

Two sum problem

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The "Two sum" problem :
Given an array of integers and a target, return the indices of the two numbers that add up to the target.

Restate the problem

- So I'm given a list of integers and a target value.
- I need to return the indices — not the values — of the two elements that sum to the target.
- I can assume exactly one solution exists."

Identify edge cases

- What if the array is empty? What if it has one element
- Can the same element be used twice
- What if there are duplicates?

Brute force first

- The naive approach: two nested loops, compare every pair — that's $O(n^2)$ time, $O(1)$ space.
- Let me describe it without coding it yet

Optimize

- Can a hash map help? Yes — I store each value and its index as I iterate.
- For each new element, I check if target - current is already in the map. That's $O(n)$ time, $O(n)$ space

Code, test, name complexity

- Write the optimized version
- trace through [2, 7, 11, 15], target=9 by hand, then name complexity.

Interviews

What interviewers assess in general.

Communication

- Do you think outloud ?
- Do you handle edge cases before coding ?

Complexity awareness

- Can you spontaneously say "this is O(n log n) time, O(1) space" ?

Adaptability

- Can you improve your solution when prompted ?

Clean code

- Meaningful names, no magic numbers, short functions

Kahoot!

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We know the way!